

SYNOPSIS

Student Information

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- **Semester:** TYBSC CS (SEM V)
- **Project Title:** AI-Powered Resume Analyzer

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1. Introduction

An AI-Powered Resume Analyzer is an intelligent web-based application designed to automate the initial screening and evaluation of resumes submitted by job applicants. Using Natural Language Processing (NLP) techniques, the system parses text from uploaded resumes (such as PDF or DOCX formats), extracts key information, and compares it against specific Job Descriptions (JDs).

The system helps recruiters quickly identify candidate suitability by calculating a match score, extracting relevant technical and soft skills, and detecting missing qualifications. It replaces tedious manual resume screening with an automated, data-driven approach.

Overall, an AI-powered resume analyzer saves hiring teams valuable time, standardizes candidate evaluations, and provides applicants with actionable feedback to improve their resumes.

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2. Problem Statement

Traditional hiring processes require HR professionals to manually read hundreds of resumes for a single job opening. This process is time-consuming, prone to human error, subject to unconscious bias, and highly inefficient when dealing with large applicant pools.

The proposed AI-Powered Resume Analyzer aims to automate resume extraction and evaluation. By analyzing candidate resumes against targeted job roles using natural language processing, the platform generates automated match scores, skill summaries, and gap analysis reports for recruiters and job seekers.

Main Problems:

- Manual screening of hundreds of resumes takes excessive time and HR effort.
- Inconsistent evaluation criteria lead to human bias and oversight.
- Difficulty in quickly verifying if a candidate possesses essential skills listed in job descriptions.
- Applicants lack objective feedback on how well their resumes match market requirements.
- Managing and organizing unstructured text data from various resume file formats is difficult.

Goal: To build an intelligent, fast, and user-friendly web application that accurately parses resumes, extracts skills, calculates job match scores, and provides structured feedback using NLP techniques.

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3. Objective of the Project

The main objective of the AI-Powered Resume Analyzer is to streamline and automate candidate screening using machine learning and NLP techniques.

Key Objectives:

- To parse unstructured text data accurately from multiple resume file formats (PDF, DOCX).
- To extract essential candidate information, such as skills, education, work experience, and contact details.
- To compare resume content with job descriptions using text similarity algorithms to compute a job match percentage.
- To highlight missing skills and key keywords needed for specific job roles.
- To provide an intuitive, interactive dashboard for recruiters and job applicants.
- To improve data security and ensure standardized, unbiased evaluation of applicants.
- To generate detailed candidate summary reports for fast decision-making.

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4. Scope of the Project

The AI-Powered Resume Analyzer is designed to process, analyze, and grade candidate resumes automatically. The system focuses on natural language extraction, skill categorization, and match scoring.

Scope Includes:

- File Upload & Parsing: Support for uploading resumes in PDF and Word formats with automatic text extraction.
- Information & Skill Extraction: Categorization of candidate skills (programming languages, tools, soft skills) and educational history.
- Job Description Matching: Algorithmic comparison between candidate skills and required job skills to produce a match percentage score.
- Keyword & Skill Gap Analysis: Identification of missing keywords and skills critical for the target job role.
- Resume Optimization Suggestions: Recommendations for job seekers to enhance their resume visibility.
- Recruiter Dashboard: A consolidated view where recruiters can upload multiple resumes, view candidate rankings, and filter applicants.
- Report Generation: Exportable summary reports containing applicant match scores and key details.

Overall Scope: The project provides an efficient, intelligent system for parsing resumes and assessing job suitability while reducing recruitment overhead.

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5. Technology Stack

- Frontend: HTML, CSS, JavaScript (or Streamlit / React.js)

- Backend: Python (Flask / FastAPI)
- NLP & AI Libraries: NLTK, spaCy, scikit-learn, PyPDF2 / pdfplumber
- Database: MySQL / SQLite
- IDE: VS Code

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6. Key Features

1. Document Upload & Parsing: Extract structured text from uploaded PDF/DOCX files.
2. Entity Recognition & Extraction: Identify names, contact details, education, and work history.
3. Skill Classification: Categorize technical skills, soft skills, and domain tools.
4. Job Match Scoring: Calculate similarity index/match score against target job descriptions using NLP techniques.
5. Skill Gap Analysis: Highlight missing keywords and skills required for specific roles.
6. Resume Score & Feedback: Provide improvement suggestions to optimize resumes for Applicant Tracking Systems (ATS).
7. Multi-Resume Comparison: Rank multiple candidates side-by-side based on match scores.
8. Exportable Analysis Reports: Generate downloadable summary reports in PDF/CSV formats.
9. Search & Filter: Search candidates by skills, match percentage, or experience level.
10. User Authentication: Secure login for recruiters and applicants.
11. Data Validation: Ensure valid file types and formats are uploaded.
12. Database Storage: Store analyzed records and user query history efficiently.

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7. Hardware and Software Requirements

- Intel Core i3 / i5 processor
- 8 GB RAM (recommended for NLP model processing)
- 20 GB+ free storage
- Python 3.9+
- Flask / FastAPI
- spaCy / NLTK / Scikit-learn
- MySQL / SQLite
- VS Code

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8. Conclusion

The AI-Powered Resume Analyzer offers a modern, automated solution to overcome the inefficiencies of manual resume screening. By using Natural Language Processing and text similarity algorithms, the application effectively extracts candidate details, matches resumes with job criteria, and highlights skill gaps.

The system reduces time-to-hire, minimizes human evaluation bias, and provides job seekers with clear insights into how to improve their resumes. Overall, this project

demonstrates how artificial intelligence and natural language processing can streamline recruitment operations and make candidate evaluation faster, objective, and accurate.